



Traumatic Stress Interacts With Bipolar Disorder Genetic Risk to Increase Risk for Suicide Attempts

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Objective: Bipolar disorder (BD) is one of the most heritable psychiatric conditions and is associated with high suicide risk. To explore the reasons for this link, this study examined the interaction between traumatic stress and BD polygenic risk score in relation to suicidal ideation, suicide attempt, and nonsuicidal self-injury (NSSI) in adolescent and young adult offspring and relatives of persons with BD (BD-relatives) compared with adolescent and young adult offspring of individuals without psychiatric disorders (controls).

Method: Data were collected from 4 sites in the United States and 1 site in Australia from 2006 through 2012. Generalized estimating equation models were used to compare rates of ideation, attempts, and NSSI between BD-relatives ($n = 307$) and controls ($n = 166$) and to determine the contribution of demographic factors, traumatic stress exposure, lifetime mood or substance (alcohol/drug) use disorders, and BD polygenic risk score.

Results: After adjusting for demographic characteristics and mood and substance use disorders, BD-relatives were

at increased risk for suicidal ideation and attempts but not for NSSI. Independent of BD-relative versus control status, demographic factors, or mood and substance use disorders, exposure to trauma within the past year (including bullying, sexual abuse, and domestic violence) was associated with suicide attempts ($p = .014$), and BD polygenic risk score was marginally associated with attempts ($p = .061$). Importantly, the interaction between BD polygenic risk score and traumatic event exposures was significantly associated with attempts, independent of demographics, relative versus control status, and mood and substance use disorders ($p = .041$).

Conclusion: BD-relatives are at increased risk for suicide attempts and ideation, especially if they are exposed to trauma and have evidence of increased genetic vulnerability.

Key words: bipolar disorder, suicide, attempted, populations at risk, polygenic risk, traumatic stress

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Individuals with bipolar disorder (BD) are at high risk for suicide attempts and suicide death.¹ As many as 20% to 25% of individuals with a clinical diagnosis of BD die by suicide.^{2,3} A systematic review by Tondo *et al.*⁴ reported that in 101 published studies from 22 countries ($N = 80,000$), 31% of those with BD made at least 1 lifetime suicide attempt versus 4% of community-residing adults. Simon *et al.*⁵ noted the relatively low ratio of suicide attempt to suicide death for persons with BD compared with the general population, which suggests stronger suicide intent and greater lethality of methods.

Although an earlier onset of BD is associated with increased risk of suicidal behaviors,⁶ ideation and attempts remain significantly under-researched in adolescents and young adults at high familial risk for BD, particularly in genetically informative samples with prospective longitudinal designs. Two studies of youth at risk for BD have reported higher rates of ideation and attempts. Klimes-Dougan *et al.*⁷ found that BD offspring at the average age of 22 years were at higher risk for suicide plans and attempts than offspring of parents with major depressive disorder and offspring of well parents; however, that study had a small sample of 37 BD offspring. Goldstein *et al.*,⁸ using data from the Pittsburgh Bipolar Offspring Study (BIOS)⁹ of 388 offspring (6–18 years of age) of 233 parents with bipolar I (BP-I) or bipolar II (BP-II) and 250 offspring of matched control parents, found that BD offspring had significantly higher rates of ideation (33% versus 20%; odds ratio [OR] 2.1).

Studies have found that as many as 50% of adults with BD report a history of childhood abuse,^{10–14} and many studies



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have reported that adults with BD who experienced child physical or sexual abuse were younger at BD symptom onset and experienced a worse clinical course compared with those without abuse.^{10-12,15} In a systematic review of childhood maltreatment in BD, Agnew-Blais and Danese¹⁶ found that maltreatment was associated with several negative clinical outcomes, including suicide attempts, and might be an early indicator of a particularly unfavorable clinical course.

The stress-diathesis model of suicidal behavior¹⁷ describes suicidal behaviors to be the result of an interaction between state-dependent stressors and a trait-like susceptibility to suicidal behavior that includes familial and genetic factors. Indeed, offspring of parents with BD might have biological vulnerabilities that increase their risk for depression and suicide,¹⁸ but also might experience disruptive home environments including exposure to parental suicidal behaviors.^{19,20}

Different studies have attempted to map specific genetic variants underlying suicidal behaviors using approaches within and across psychiatric diagnoses, so far with limited success (reviewed by Sokolowski *et al.*²¹). This is largely due to the relatively small samples hitherto examined (typically <5,000 individuals with suicide attempt data). However, as with the genetic architecture of psychiatric diagnoses,^{22,23} susceptibility to ideation and attempt appears to be highly polygenic.²⁴⁻²⁶ In one of the larger genome-wide association studies to date, Willour *et al.*²⁴ examined 1,200 individuals with BD who had attempted suicide and 1,497 BD non-attempters, finding no single nucleotide polymorphisms (SNPs) exceeding genome-wide significance for association. However, of 2,507 SNPs with a *p* value less than .001 in the primary study, 49 replicated to a *p* value less than .01 with the same direction of effect in an independent cohort of 1,295 attempters with BD and 1,822 non-attempters with BD.²⁴ The genetic overlap between risk genes for psychiatric disorders and suicide attempts is unclear, although preliminary studies in small samples have suggested partial pleiotropy.²⁵ Polygenic risk scores (PRSs) derived from the sum of individual effects of many hundreds of BD-associated variants could help elucidate the relative contributions of genes versus environment for suicidal behaviors in BD.

In the present study, we examined suicidal ideation, attempts, and nonsuicidal self-injury (NSSI) in 307 adolescent offspring and relatives of adults affected with BD (BD-relatives) compared with 166 offspring and relatives of adults without specific psychiatric disorders (controls). The aims of the study were to assess whether BD-relatives are at greater risk of suicidal or self-harm behaviors than controls, independent of a personal history of mood or substance use disorders; whether an individual's genetic risk for BD (i.e., polygenic risk score) is indicative of suicidal or self-harm behaviors; and whether genetic risk for BD interacts with traumatic event exposures to increase the risk for suicidal or self-harm behaviors, independent of the presence of a mood disorder or substance use disorders.

METHOD

Participants 12 to 21 years old were ascertained from 5 independent sites, 4 in the United States (Johns Hopkins University, University of Michigan, Washington University in St. Louis, and

Indiana University²⁷) and 1 in Australia (University of New South Wales).²⁸ All clinical procedures were approved by the institutional review boards at the 5 centers. Written informed consent (or assent with parental consent for US participants <18 years old and Australian participants <17 years old) was obtained after a thorough explanation of the study. Participants were recruited using a "top-down" ascertainment method: that is, all offspring or other relatives in multiplex families of a proband with a confirmed *DSM-IV* diagnosis of BP-I, BP-II, or schizoaffective disorder bipolar type (SAB) who were 12 to 21 years old were eligible for inclusion, independent of the diagnostic status of the offspring. *DSM-IV* diagnosis was derived through the Diagnostic Interview for Genetic Studies (DIGS)²⁹ and the Family Instrument for Genetic Studies (FIGS).³⁰ BD-relatives were predominantly (88%) children of a proband with BD ascertained through the National Institute of Mental Health Genetics Initiative bipolar sample or similar genetic studies³¹⁻³³; only 6% were siblings of a proband with BD, and 6% were second-degree relatives of a proband from multiplex families in which a first-degree relative with BP-I or SAB also existed. Control participants were in the same age range but had no first-degree relative with a *DSM-IV* diagnosis of BP-I or BP-II, SAB, recurrent major depression, schizophrenia, recurrent substance abuse, or any past psychiatric hospitalization; and no parent with a first-degree relative who had a past mood disorder hospitalization or history of psychosis. Controls were recruited through general medical clinics, motor vehicle records, print and electronic media, and notice boards in universities and local communities. BD-relatives came from 197 families: 116 with a single offspring, 60 with 2 offspring, 16 with 3 offspring, 3 with 4 offspring, and 1 each with 5 and 6 offspring. Controls were from 120 families: 86 with a single offspring, 24 with 2 offspring, 9 with 3 offspring, and 1 with 5 offspring.

Data were collected from June 1, 2006 through September 30, 2012. In families of BD-relatives and controls, all offspring were interviewed using the Schedule for Affective Disorders for School-Age Children (K-SADS)³⁴ and provided a blood specimen for DNA analysis. At least 1 parent had to be available for interview; parents were interviewed about themselves (using the DIGS), about their spouse and other relatives (using the FIGS), and about all offspring (using the K-SADS-Parent Version).

Because existing versions of the K-SADS did not define specific episodes in time and duration before assessing symptoms and did not include questions targeting each *DSM-IV* criterion item for affective disorders, we adapted questions and interviewer instructions from the DIGS to address these issues. The fully computerized K-SADS for BD is available at https://www.nimhgenetics.org/interviews/k-sads_bp_study55/index.php. Diagnoses were generated by best-estimate clinical consensus. Interviewers were extensively trained by the principal investigators and coordinators at each site after a weeklong common training at the Indiana University coordinating site. At each site, the interviewers or clinical coordinators had an extensive clinical background. Interviews were performed in person or by telephone. The reliability of telephone interview for such purposes has been documented.³⁵ Clinicians performing the best-estimate diagnoses were not blinded to the BD-relative versus control status of the participant³⁶ but were blinded to the specific hypotheses of the study. Diagnoses and age at onset determinations were made based on consensus between 2 clinicians (i.e., psychiatrists, clinical psychologists, or clinical social workers), including information from the direct interview, medical records, and parent interview. Lifetime diagnoses were assigned according to *DSM-IV* criteria, including BD not otherwise specified, which was generally diagnosed only if the participant approached the criteria for BP-II but had 1 fewer symptom in the hypomanic and depressive categories.

Measures

Dependent Variables. Lifetime occurrence of suicide ideation, of suicide attempts, and of NSSI was studied independently over the course of follow-up using data from all available K-SADS diagnostic assessments. NSSI is defined as the direct and deliberate infliction of pain or tissue damage without suicidal intent.³⁷ Most lifetime ideations, attempts, and NSSIs were reported at baseline at a mean age of 17 years. From baseline to the end of follow-up, there were 17 additional incident reports of ideation, 7 incident reports of attempts, and 10 incident reports of NSSI.

Independent Variables, Psychiatric Diagnoses. Lifetime DSM-IV mood disorders (major depression and BD) and a variable including lifetime DSM-IV substance use disorder (i.e., alcohol abuse or dependence and drug abuse or dependence) were computed using best-estimate diagnoses from the K-SADS.

Independent Variables, Traumatic Events. Traumatic event exposures were assessed through the Stressful Life Events Schedule (SLES) Child-Reported Version,³⁸ which assesses the presence of more than 80 possible stressors in the past 12 months. Three SLES variables were used to tap exposure to bullying, domestic violence, and sexual abuse—each a known risk factor or correlate of ideation and attempt. Endorsement of any of the 3 SLES traumatic event exposures were computed into a fourth composite variable, named “any trauma history.”

Genotyping and Quality Control. Peripheral blood samples were collected from all individuals for genetic analysis. DNA was extracted from whole blood by the Rutgers University Cell and DNA Repository (New Brunswick, NJ; US participants) or Genetic Repositories Australia (Sydney, NSW, Australia; Australian participants), as previously described.³⁹ Genome-wide SNP genotyping was conducted using the Infinium PsychArray BeadChip (Illumina, Scoresby, VIC, Australia) at the Mt. Sinai School of Medicine Genomics Core Facility, with genotype calling and quality control conducted using standard Psychiatric Genomics Consortium (PGC) pipelines. In brief, genotypes were called from 3 algorithms (birdseed, zCall, genCall) and SNPs with Hardy-Weinberg equilibrium (HWE) $p < 1e-06$ or genotype call rates lower than 97% were removed, and the resulting files were merged in PLINK.⁴⁰ A second quality control was applied to remove individuals with SNP call rates lower than 98%, individuals with an inbreeding coefficient outside ± 0.2 , or genotype-derived sex discrepancy with recorded sex from clinical files. SNPs with minor allele frequency lower than 0.00001 and those with HWE $p < 1e-06$ in controls or HWE $p < 1e-10$ in cases were excluded, leaving 426,091 SNPs. The successfully genotyped SNPs had a higher than 99.6% genotype pass rate.

Imputation and multidimensional scaling analysis to develop genotype-derived ethnicity principal components was conducted according to ENIGMA2 protocols (<http://enigma.ini.usc.edu/>). Briefly, post-QC genotype files were merged with genotypes from ethnically diverse 1,000 genome samples (Human Mapping Set, phase 1 release 3; Thermo Fisher Scientific, Waltham, MA) using SNPs that are polymorphic in European populations. SNPs with minor allele frequency lower than 0.01, genotype call rate lower than 95%, strand ambiguity, and those with HWE $p < 1e-06$ were excluded before haplotype phasing and imputation using mach and minimach.^{41,42}

Multidimensional scaling analysis was conducted in PLINK using 164,680 independent SNPs across the panel, from which principal components were calculated. Principal components values from this sample were compared with those of participants of known ancestry to derive a genotype-derived ancestry allocation for each participant. Identity-by-descent estimates were generated from 75,182 independent SNPs and were used to confirm documented

familial relationships, identify any spurious relationships between participants, and identify gender discrepancies between genotype-derived and documented gender, resulting in subject exclusion.

PRS Determination. BD-associated SNPs were selected based on prior evidence of genetic association from the PGC1-BD discovery sample.²³ PRSs were computed from index SNPs after clumping based on linkage disequilibrium, using the *score* function in PLINK,⁴⁰ and were weighted by the logarithm of the odds ratio of the risk allele from the original discovery genome-wide association study.²³ Before computing BD-PRSs, strand-ambiguous SNPs, insertion/deletion variants, and those with imputation r^2 less than 0.8 were excluded. BD-PRS was tested at 3 increasingly liberal p value thresholds: the first representing SNPs whose discovery p value was less than .0001 (BD-PRS $p < .0001$; $n = 97$ SNPs), the second representing SNPs whose discovery p value was less than .001 (BD-PRS $p < .001$; $n = 97 + 440$ SNPs), and the third representing SNPs whose discovery p value was less than .01 (BD-PRS $p < .01$; $n = 3,011$ SNPs; $97 + 440 + 2,474$ SNPs). Participants were dichotomized into a high or low BD-PRS category for statistical analysis, where a high BD-PRS represents the top 2 quintiles and a low BD-PRS represents the bottom 3 quintiles.

Demographic Characteristics

Age, sex, and ethnicity were determined at baseline interview. Self-reported ethnicity, using the 7 US Census categories, was consistent with genotype-derived ethnicity (98.7% concordance). Based on genotype-derived ethnicity, participants were categorized as Caucasian, African, or other. Caucasian and African groups were used as covariates in the analyses, and each was compared with the other ethnic group. To explore whether the findings were subject to confounding by ethnicity, we conducted additional analyses in Caucasians only.

Statistical Analyses

Demographic and clinical characteristics were compared between BD-relatives ($n = 307$) and controls ($n = 166$) and between those with and without suicidal ideation, attempts, and NSSI using generalized estimating equation models^{43,44} implemented in R software (<https://www.r-project.org/>). Generalized estimating equation modeling enables specifying a binomial distribution for the outcome and an exchangeable clustering pattern within a family to correct for non-independence of measurement between siblings. This modeling approach also was implemented to test whether a high BD-PRS is predictive of ideation, attempts, and NSSI and whether it interacts with traumatic event exposures, adjusting for BD-relative status, ethnicity, and the presence of DSM-IV mood disorders and substance use disorders. Interactions between a BD-PRS high or low category and trauma exposure were estimated for models with significant main effects. All models using the BD-PRS included adjustment for genetically informed ethnicity variables.

RESULTS

The demographic and clinical characteristics of the BD-relative and control groups are presented in Table 1. The BD-relative and control groups were gender balanced and were mainly of European ancestry (85% versus 65% for BD-relative versus control, $p < .001$). At the time of the baseline interview, the mean age of BD-relatives was slightly younger than that of the control group (16.7 versus 17.1 years, $p = .029$). As expected, BD-relatives were more likely

TABLE 1 Demographic and Clinical Characteristics of the Sample

	BD Relatives (n = 307)		Controls (n = 166)		OR	95% CI	p Value
	n	%	n	%			
Demographics							
Girls	152	50	79	48	0.9	0.9–1.1	.929
Caucasian	261	85	108	65	2.8	1.5–5.3	.001
Age at baseline, mean (SD)	16.7	3.0	17.1	2.7	0.9	0.9–0.9	.029
Axis I diagnoses							
Mood disorder	152	50	36	22	1.4	1.2–1.6	<.001
Substance disorder	91	30	29	17	1.2	1.1–1.3	.006
Conduct disorder	13	5	1	0.01	1.4	1.3–1.8	.0156
ADHD	17	6.5	4	03	1.1	0.95–1.3	.191
Self-injurious thoughts/behaviors							
Ideation	72	23	15	9	1.3	1.1–1.5	<.001
Attempt	34	11	6	4	1.3	1.1–1.5	.003
NSSI	55	18	22	13	1.1	0.9–1.2	.192
Trauma history							
Domestic violence	24	8	5	3	0.9	0.8–1.0	.217
Sexual abuse	12	4	4	2	0.9	0.8–1.0	.176
Bullied	45	15	15	9	0.9	0.9–1.0	.324
Any trauma history	69	22	23	14	1.0	0.9–1.1	.657

Note: Statistics include no covariates. Bold text indicates statistical significance at $p < .05$. ADHD = attention-deficit/hyperactivity disorder; BD = bipolar disorder; NSSI = nonsuicidal self-injury; OR = odds ratio.

to have a *DSM-IV* lifetime mood disorder (OR 1.4, 95% CI 1.2–1.6, $p < .001$) and lifetime substance use disorder (OR 1.2, 95% CI 1.1–1.3, $p = .006$).

BD-relatives were 30% more likely to have ideation and attempts than controls (OR 1.3, 95% CI 1.1–1.5, $p < .001$; OR 1.3, 95% CI 1.1–1.5, $p = .003$; respectively) but were not more likely to have NSSI (OR 1.1, 95% CI 0.9–1.2, $p = .192$). There were no group differences between BD-relatives and controls in exposure to domestic violence, sexual abuse, being bullied, or a composite measure of any traumatic event exposure.

After adjusting for offspring mood and substance use disorders and demographic factors in the full sample, ideation and attempts were significantly more prevalent in BD-relatives compared with controls (OR 1.1, 95% CI 1.0–1.1, $p = .012$; OR 1.1, 95% CI 1.0–1.2, $p = .011$) (Table 2). NSSI did not differ between BD-relatives and controls. In the sample of Caucasian-only participants, ideation and

attempts were more common in BD-relatives than in controls after adjusting for demographic variables (model 2), but did not remain significant after adjusting for offspring mood and substance use disorders (data not shown).

Table 3 presents results of generalized estimating equation analyses comparing those with and without ideation, attempts, and NSSI. Those with and without ideation, attempts, and NSSI did not differ according to sex or baseline age. Caucasian participants were less likely to report ideation, attempts, and NSSI. Those with ideation were more likely to have a mood disorder than those without (OR 17.7, 95% CI 8.3–37.7, $p < .001$), as were attempters versus non-attempters (OR 12.1, 95% CI 4.1–35.3, $p < .001$) and those with NSSI versus those without NSSI (OR 20.5, 95% CI 9.3–45.2, $p < .001$). Those with ideation were more likely than those without ideation to have a substance use disorder (OR 3.0, 95% CI 1.7–5.3, $p < .001$), as were attempters versus non-attempters (OR 4.9, 95% CI 2.2–10.8, $p < .001$) and those

TABLE 2 Unadjusted and Adjusted Generalized Estimating Equations Analyses of Relative Status on Lifetime Ideation, Attempt, and Nonsuicidal Self-Injury (NSSI)

All Participants	Model 1			Model 2			Model 3		
	OR	95% CI	p Value	OR	95% CI	p Value	OR	95% CI	p Value
Ideation	1.3	1.1–1.5	<.001	1.2	1.1–1.3	<.001	1.1	1.0–1.1	.012
Attempts	1.3	1.1–1.5	.003	1.3	1.1–1.6	<.001	1.1	1.0–1.2	.011
NSSI	1.1	0.9–1.2	.192	1.1	1.0–1.2	.016	1.0	0.9–1.1	.173

Note: Results in boldface are statistically significant. Model 1 is unadjusted; 307 individuals with relative with bipolar disorder of 473 individuals. Model 2 adjusted for demographics (age, sex, and ethnicity); 286 individuals with relatives with bipolar disorder of 432 individuals. Model 3 adjusted for demographics, mood disorders, and substance use disorders; 283 individuals with relatives with bipolar disorder of 422 individuals. OR = odds ratio.

TABLE 3 Characteristics of Those With Suicidal Ideation, Attempt, and Nonsuicidal Self-Injury (NSSI)

	Ideation ^a			Attempt ^b			NSSI ^c		
	OR	95% CI	p Value	OR	95% CI	p Value	OR	95% CI	p Value
Demographics									
Female sex	0.9	0.5–1.7	.775	0.9	0.4–1.9	.729	1.3	0.7–2.5	.392
Caucasian	0.3	0.1–0.9	.029	0.1	0.2–0.4	.001	0.3	0.1–0.9	.040
Age at baseline	1.0	0.9–1.1	.918	1.0	0.8–1.1	.515	1.1	0.9–1.2	.105
Axis I diagnoses									
Mood disorder	17.7	8.3–37.7	<.001	12.1	4.1–35.3	<.001	20.5	9.3–45.2	<.001
Substance use disorder	3.0	1.7–5.3	<.001	4.9	2.2–10.8	<.001	2.8	1.5–5.0	<.001
Traumatic stress history									
Domestic violence	0.9	0.6–1.4	.550	1.3	0.7–2.4	.379	0.9	0.5–1.3	.408
Sexual abuse	0.9	0.5–1.4	.597	1.5	0.8–2.8	.163	0.9	0.6–1.4	.642
Bullied	1.2	0.8–1.8	.394	1.4	0.8–2.6	.231	0.9	0.6–1.4	.701
Any trauma history	1.3	0.8–2.0	.252	2.1	1.2–3.8	.014	0.9	0.6–1.4	.743

Note: Results in boldface are statistically significant. Generalized estimating equations analyses adjusted by demographics, mood and substance use disorders, and past-year traumatic stress history. OR = odds ratio.

^an = 87 vs. 336 with no ideation, attempt, or NSSI.

^bn = 40 vs. 336 with no ideation, attempt, or NSSI.

^cn = 77 vs. 336 with no ideation, attempt, or NSSI.

with NSSI versus those without NSSI (OR 2.8, 95% CI 1.5–5.0, $p < .001$). The only trauma history construct that was associated with any of the 3 suicide behavior outcomes was that any trauma history was 2-fold higher in suicide attempters than in non-attempters (OR 2.1, 95% CI 1.2–3.8, $p = .014$).

Using the SNPs most robustly associated with BD from the PGC, we conducted analyses to determine whether a BD-PRS was higher in those with suicidal or self-harm behaviors compared with those with no suicidal or self-harm behaviors. Of participants of all ethnic backgrounds, BD-relatives were more likely to have a higher BD-PRS than controls at the $p < .001$ and $p < .01$ thresholds (BD-PRS $p < .001$, OR 1.1, 95% CI 1.0–1.2, $p = .030$; and BD-PRS $p < .01$, OR 1.2, 95% CI 1.1–1.3, $p = .001$) after adjusting for age, sex, principal component analysis ethnicity, mood and substance use disorders (Table 4).

Those with higher BD-PRS at $p < .01$ were marginally more likely to have attempted suicide after adjusting for BD-relative versus control group, age, sex, principal component analysis ethnicity, and mood and substance use disorders (OR 2.3, 95% CI 1.0–5.4, $p = .061$). However, the mean BD-PRS did not differ between those with and without ideation or NSSI at any p value threshold.

Because the PGC discovery cohort was largely ethnically Caucasian, computation of BD-PRS for ethnically diverse participants could have resulted in distortion of the PRS owing to ethnically heterogeneous odds ratios and allele frequencies. Therefore, we repeated all BD-PRS analyses in Caucasian-only participants, representing approximately 80% of the full sample. BD-relatives were more likely to have a higher BD-PRS than controls at $p < .001$ and $p < .01$ (OR 1.2, 95% CI 1.0–1.4, $p = .035$; and OR 1.2, 95% CI 1.0–1.5, $p = .039$; respectively) after adjusting for BD-relative versus

TABLE 4 Bipolar Disorder Polygenic Risk Score (PRS) Associations

	PRS p Value Threshold < .0001			PRS p Value Threshold < .001			PRS p Value Threshold < .01		
	OR	95% CI	p Value	OR	95% CI	p Value	OR	95% CI	p Value
All participants(n/N)									
Ideation (79/386)	1.0	0.5–1.8	.914	0.9	0.5–1.6	.720	1.3	0.7–2.4	.367
Attempt (35/342)	0.5	0.2–1.3	.169	1.2	0.5–2.6	.669	2.3	1.0–5.4	.061
NSSI (75/382)	1.0	0.5–1.9	.943	1.1	0.6–2.0	.788	1.3	0.7–2.5	.420
Relative status (283/431)	1.0	0.9–1.1	.275	1.1	1.0–1.2	.030	1.2	1.1–1.3	.001
Caucasian-only									
Ideation (68/328)	1.1	0.6–2.2	.745	0.9	0.5–1.6	.671	1.1	0.6–2.1	.714
Attempt (26/286)	0.6	0.2–1.8	.388	1.1	0.5–2.6	.787	1.5	0.6–3.4	.354
NSSI (59/319)	1.1	0.5–2.3	.775	1.0	0.5–2.0	.932	1.1	0.6–2.1	.775
Relative status (255/362)	1.0	0.9–1.2	.755	1.2	1.0–1.4	.035	1.2	1.0–1.5	.039

Note: Results in boldface are statistically significant. Models for suicide/nonsuicidal self-injury (NSSI) outcomes adjusted for sex, age, ethnicity, mood disorders, substance use disorders, and relative status. OR = odds ratio.

control group, age, sex, ethnicity, mood and substance use disorders. However, in the smaller Caucasian-only sample (80% of the sample), results were not statistically significant for ideation, attempts, and NSSI (Table 4).

To evaluate interactions of BD-PRS with trauma on the risk for suicide outcomes and relative status, we chose a single p value discovery threshold less than .001. Interestingly, for those participants who experienced trauma exposure and had a high BD-PRS $p < .001$, the risk for suicide attempt was significantly increased after adjusting for sex, baseline age, BD-relative versus control group, ethnicity, and lifetime DSM-IV mood disorders and substance use disorders (OR for interaction term = 8.6, 95% CI 1.1, 68.0, $p = .041$; Table 5). Stronger effects were observed in the Caucasian-only analysis (OR for the interaction term = 9.4, 95% CI 1.1, 83.3, $p = .043$; data not shown) after adjusting for demographic characteristics, mood disorders, and substance use disorders.

DISCUSSION

This study uniquely examined suicidal and self-harm behaviors in a young cohort of individuals at increased risk of BD, although many had not yet developed BD. We found that offspring of BD-relatives reported more suicidal ideation and attempts than controls. This finding was sustained after adjusting for the presence of mood disorders and substance use disorders, suggesting that parental BD is a key correlate of suicidal ideation and attempts, and that genetic risk for suicidal behavior is not wholly acting through a genetic pathway for mood disorder. Importantly, we found that a high BD-PRS plus traumatic event exposure was associated with increased suicide attempts, independent of mood or substance use disorders and ethnicity. This could imply that trauma exposure increases the risk of attempts especially in those at genetic risk, regardless of the presence of mood disorders.

To our knowledge, no prior studies have examined the relationship between BD-PRS and suicide attempt, let alone an interaction with traumatic events. The most permissive BD-PRS (derived from SNPs at $p < .01$) marginally differentiated those with from those without attempts, independent of relationship status, ethnicity, and mood and substance use disorders, whereas the more restrictive PRS did not. This could indicate the contribution of other genetically mediated aspects common to BD and suicide behavior, such as impulsivity, aggression, or severity of social, cognitive, or psychiatric impairment. Indeed, these pleiotropic effects could be more apparent in PRSs derived from a larger number of variants, which more likely would encompass risk variants for BD and for impulsive traits. The lack of association in the Caucasian-only analysis could indicate that population stratification is driving the association with suicide attempt in the larger and more ethnically diverse sample or that it is influenced by the loss of sample size, resulting in a weaker power to detect significant differences. Because the effect sizes in the Caucasian-only analysis were similar, the smaller sample could have influenced those results; thus, replication in a larger ethnically homogeneous cohort is warranted. Identification of specific genes influencing suicide attempt and exploration of genetic pleiotropy between BD and suicide attempt will soon be possible, pending the large-scale genome-wide association analysis of suicide attempt currently underway as part of the PGC.

There is a dearth of information on the link between NSSI and BD, although the risk factors for NSSI have been reported to be similar to those for suicide attempts in young people.^{45,46} Intent to die is not evident in NSSI, and the motives behind NSSI and suicide attempts differ.⁴⁷ Small clinical studies have found an increase in NSSI in those with BD,^{48,49} yet a recent meta-analysis on risk factors for NSSI did not find an association between NSSI and BD.⁵⁰ NSSI had some

TABLE 5 Results of Polygenic Risk Score by Trauma Exposure Interaction

	No Trauma			Trauma			p Value for Interaction ^c
	OR	95% CI	p Value	OR	95% CI	p Value	
Relatives ^a							
With low PRS (152/259)	1.1	1.0–1.3	.066				
With high PRS (132/173)				1.3	0.9–1.7	.125	.545
Ideation ^b							
With low PRS (50/246)	0.78	0.3–1.3	.553				
With high PRS (30/151)				1.1	0.4–1.2	.844	.348
Attempt ^b							
With low PRS (19/205)	0.27	0.05–1.4	.111				
With high PRS (16/137)				2.7	0.9–8.3	.089	.041
NSSI ^b							
With low PRS (45/231)	0.80	0.4–1.8	.574				
With high PRS (30/151)				1.1	0.3–3.9	.883	.220

Note: Results in boldface are statistically significant. Bipolar disorder polygenic risk score (PRS) p threshold is .001. NSSI = nonsuicidal self-injury; OR = odds ratio.

^aAdjusted for demographics, mood disorders, and substance use disorders.

^bAdjusted for demographics, bipolar disorder relative status, and mood and substance use disorders.

^cInteraction p value is computed using the full sample and reflects the increased risk in the outcome for those with trauma and high PRS compared with those with no trauma and low PRS.

different correlates than suicidal ideation and attempts in our study, because NSSI was associated with sex and age at interview, but not with trauma exposure. Unlike suicidal ideation and attempts, NSSI was not significantly higher in BD-relatives than in controls in the univariate model.

We should acknowledge the limitations of the present observations. First, even in a high-risk study, suicide attempts and exposures to trauma were relatively uncommon in our sample; thus, a composite trauma exposure was used. Second, the assessment of childhood diagnoses rested on a combination of retrospective data from child and parental observations, supplemented with medical records when available. Third, we applied a genetic model that assumes additivity of risk alleles in a single aggregate score in accord with methods reported elsewhere,²² although this does not model more complex genetic interactions underlying genetic risk. Fourth, measurement of trauma and stress (SLES) was during the 12 months before baseline assessment, whereas the occurrence of suicidal attempts, suicidal ideation, and NSSI is a lifetime evaluation. The counterbalancing strengths are that this study included well-characterized at-risk youth from a genetically informative sample.

Schaffer *et al.*,⁶ as part of the International Society for Bipolar Disorders Task Force on Suicide meta-analysis, noted the importance of integrating genetic, epigenetic, and social learning mechanisms when building models of psychobiological contributors to suicidal behaviors in BD. Early-life adversity and epigenetic mechanisms seem to be related to causal mechanisms for this diathesis. For example,

SKA2 DNA methylation mediates responsivity of the hypothalamic-pituitary-adrenal axis in the context of childhood trauma⁵¹ and has been associated with a range of psychopathologies related to the hypothalamic-pituitary-adrenal axis, including posttraumatic stress disorder⁵² and suicidal behavior.^{51,53} Therefore, biomarkers might help to inform risk-assessment procedures and treatment choice for the prevention of future suicidal behaviors. &

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Clinical Guidance

- We found that offspring of individuals with BD are at risk for suicidal attempts and ideation, but not only through specific mechanisms associated with mood disorder vulnerability.
- We also found that genetic susceptibility to BD can increase the risk for suicide attempt, but only in those who also have experienced trauma.
- This work highlights the importance of severe environmental stressors in the development of suicide attempts in those at risk for BD.

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